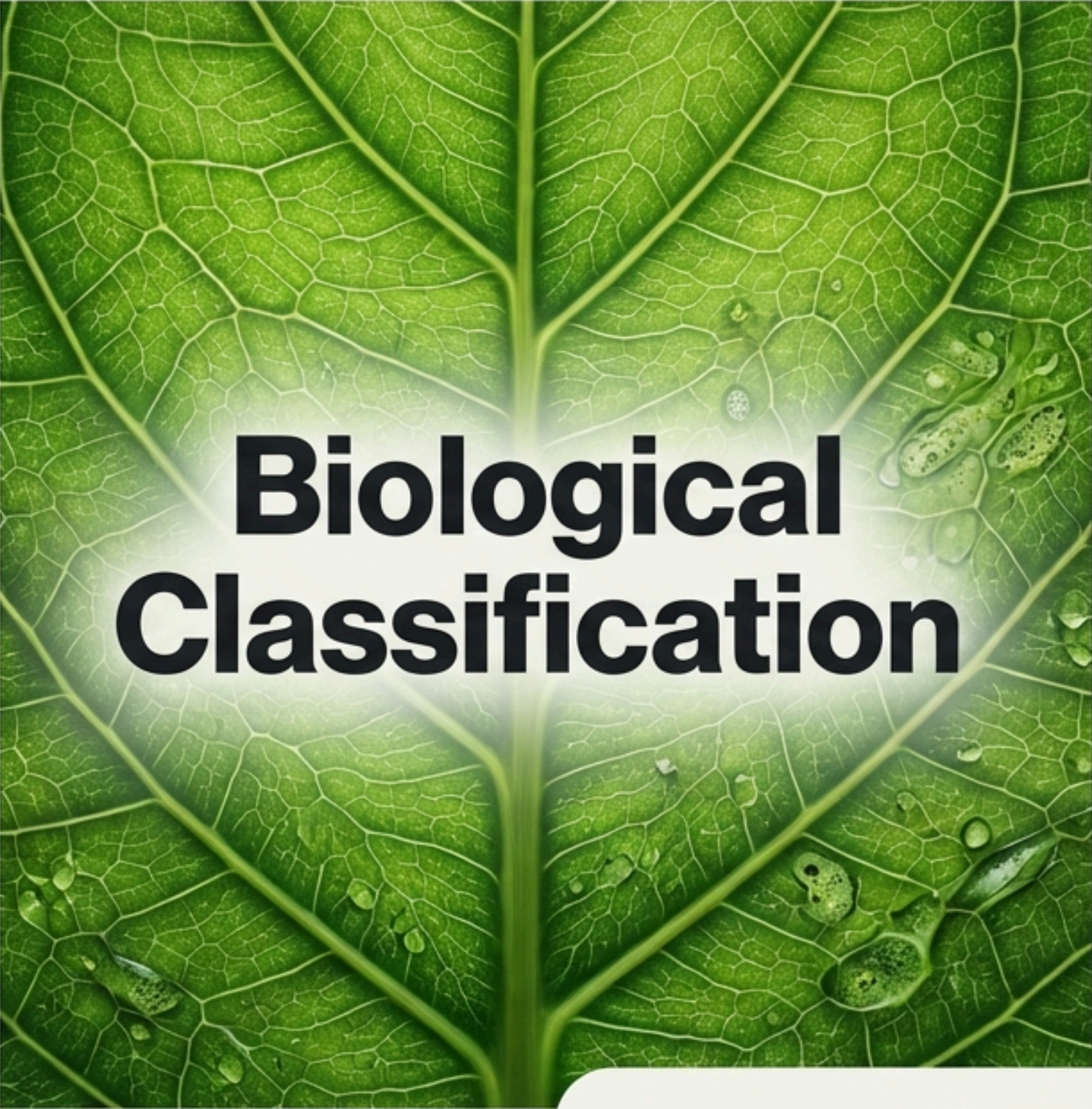




Biological Classification



The Architects & The Acellular Enigmas

————— A High-Yield Review: —————

Kingdom Plantae | Kingdom Animalia | Viruses | Viroids | Prions | Lichens

Kingdom Plantae: The Green Architects

Core Characteristics:

- **Structure:** Eukaryotic with prominent chloroplasts.
- **Cell Wall:** Composed mainly of **Cellulose**.
- **Includes:** Algae, Bryophytes, Pteridophytes, Gymnosperms, Angiosperms.

The Heterotrophic Exceptions

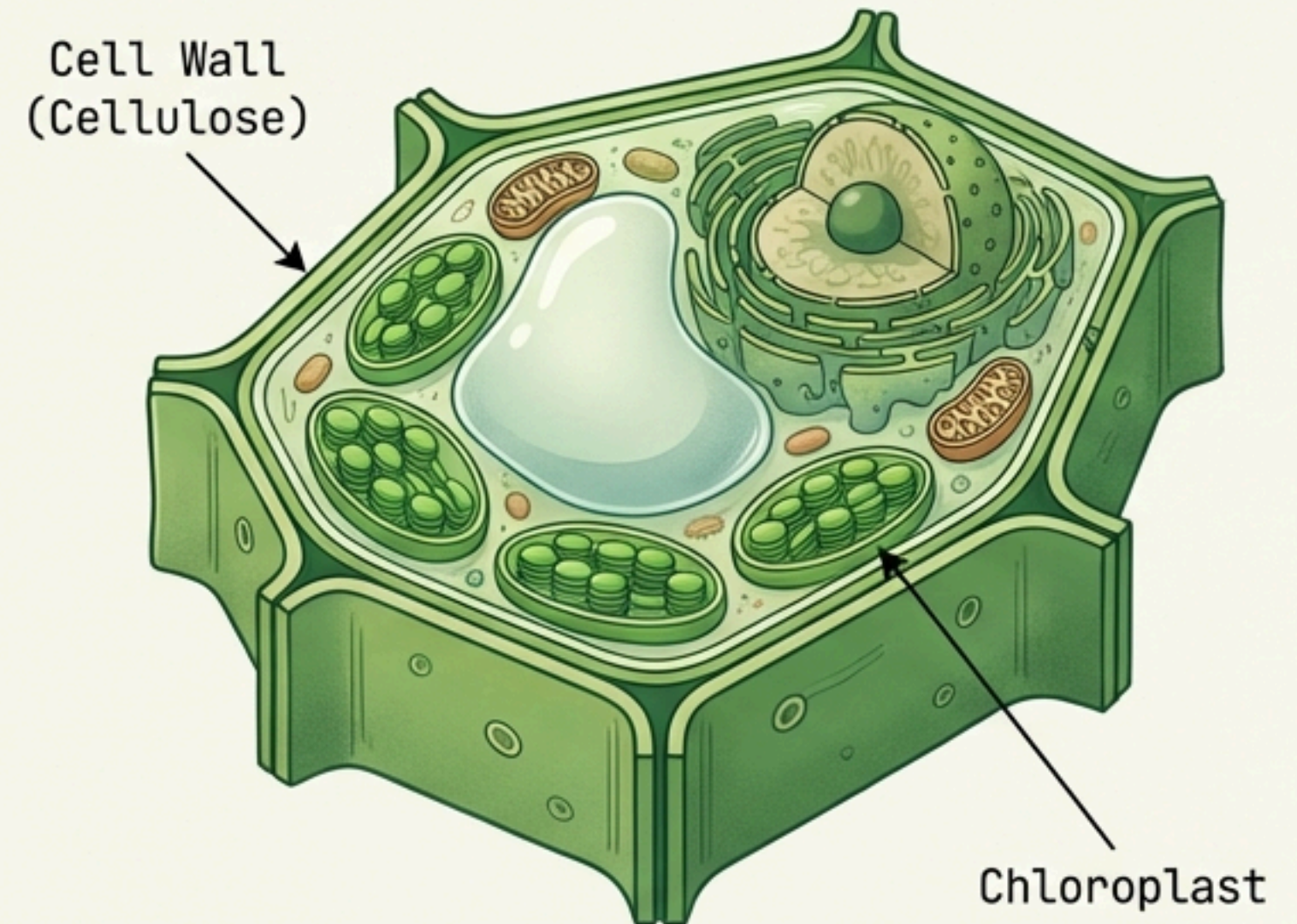
A few members are **partially heterotrophic** such as the insectivorous plants or parasites.



Insectivorous



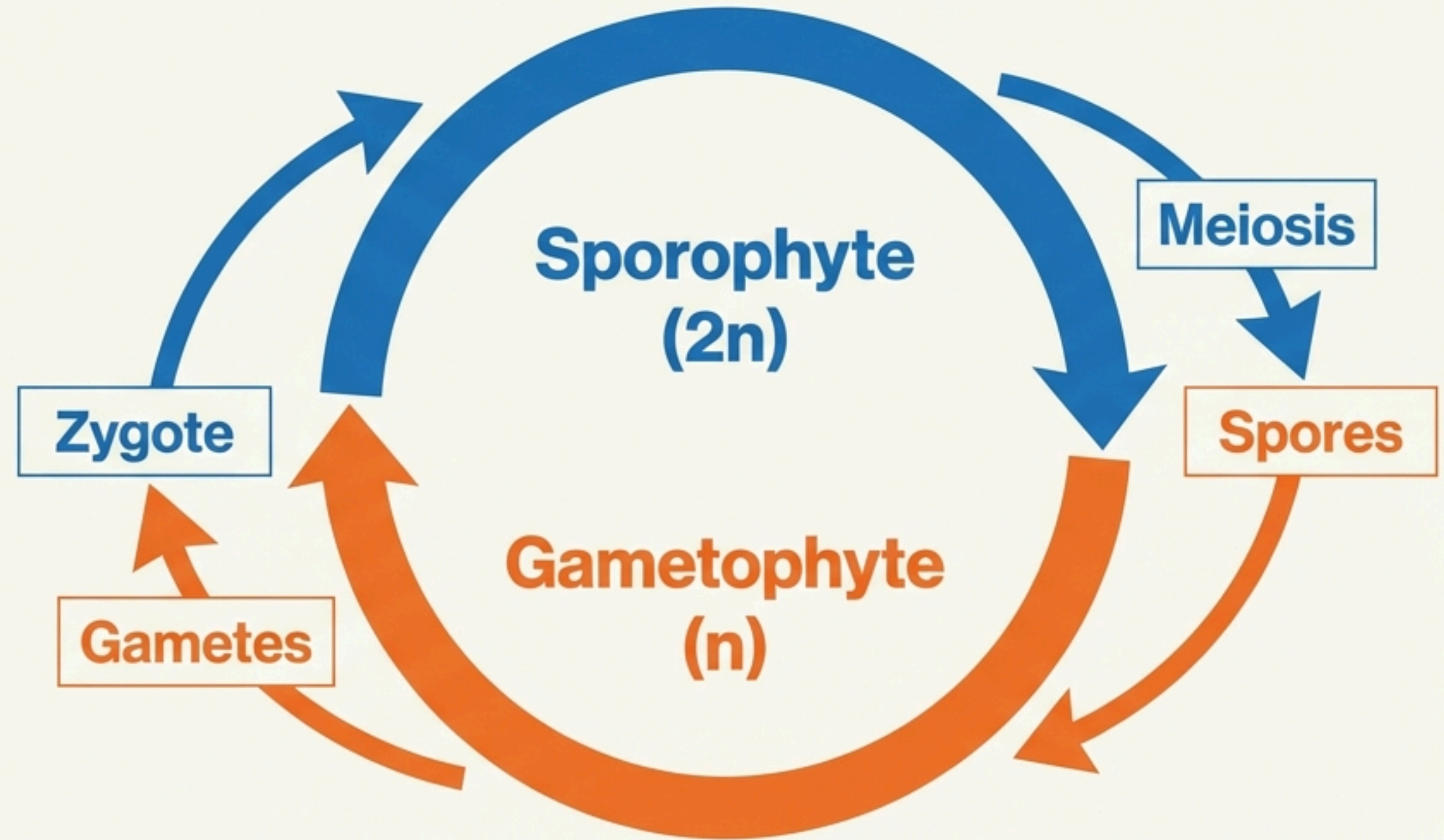
Parasite (*Cuscuta*)



Alternation of Generations

The Dual Phases:

1. Diploid Sporophytic
2. Haploid Gametophytic



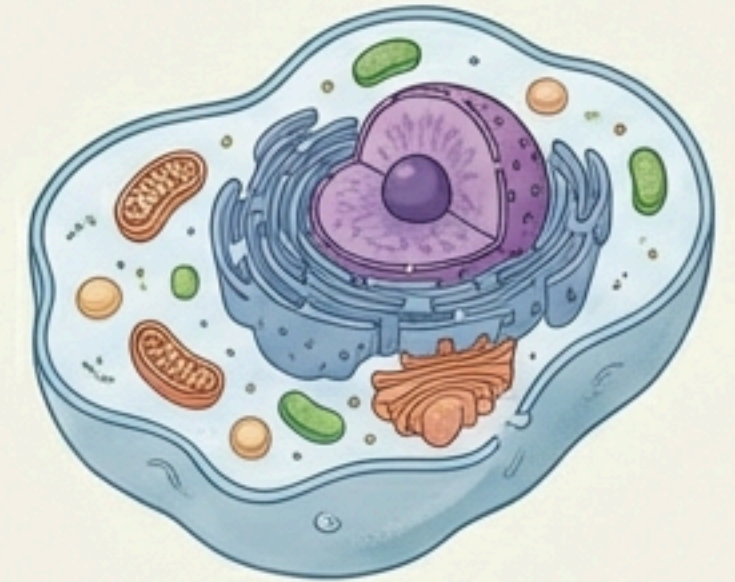
Exam Focus: The lengths of the haploid and diploid phases, and whether these phases are free-living or dependent on others, vary among different groups in plants.

Kingdom Animalia: The Heterotrophic Consumers

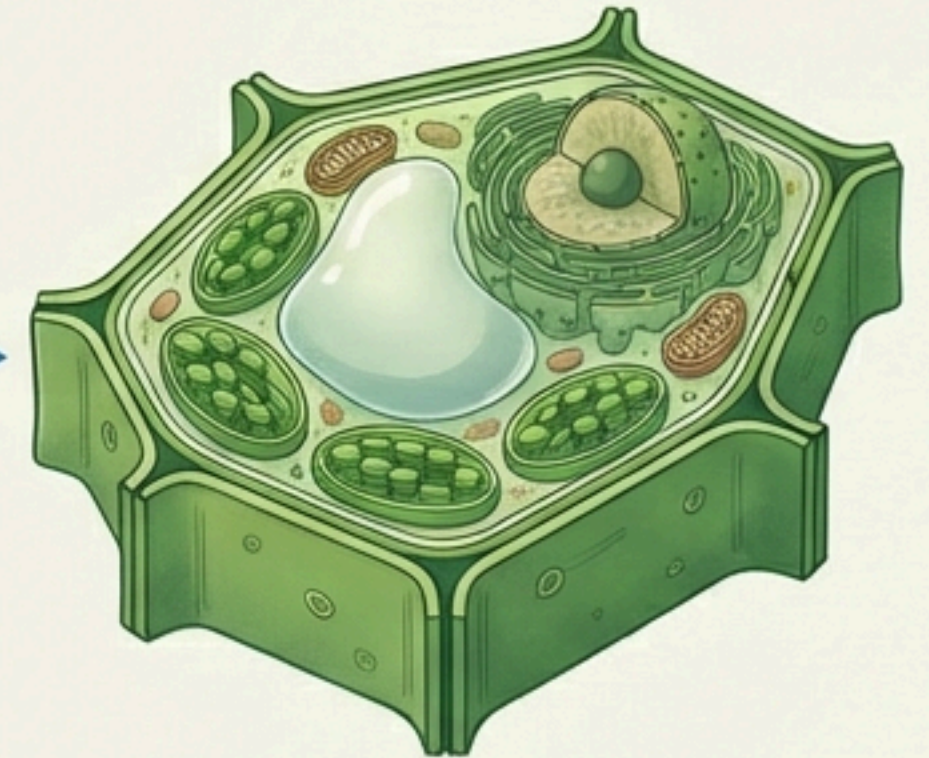
Defining Characteristics:

- Heterotrophic, Eukaryotic, Multicellular.
- **LACK** cell walls.
- Nutrition: **Holozoic** (ingestion of food).
- Digestion: Internal cavity.
- Storage: Food reserves stored as **Glycogen** or **Fat**.

No
Cell Wall



Cell Wall
Present



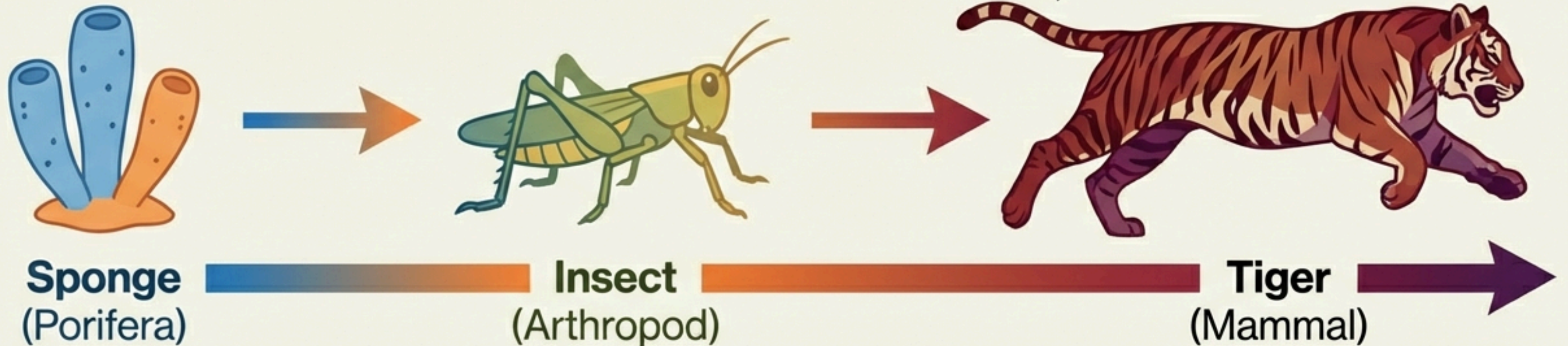
Exam Alert: They directly or indirectly depend on plants for food.

Physiology & Growth

Growth Pattern: Animals follow a definite growth pattern.

Result: Adults have a **definite shape and size**.

Higher forms show elaborate sensory and neuromotor mechanisms.



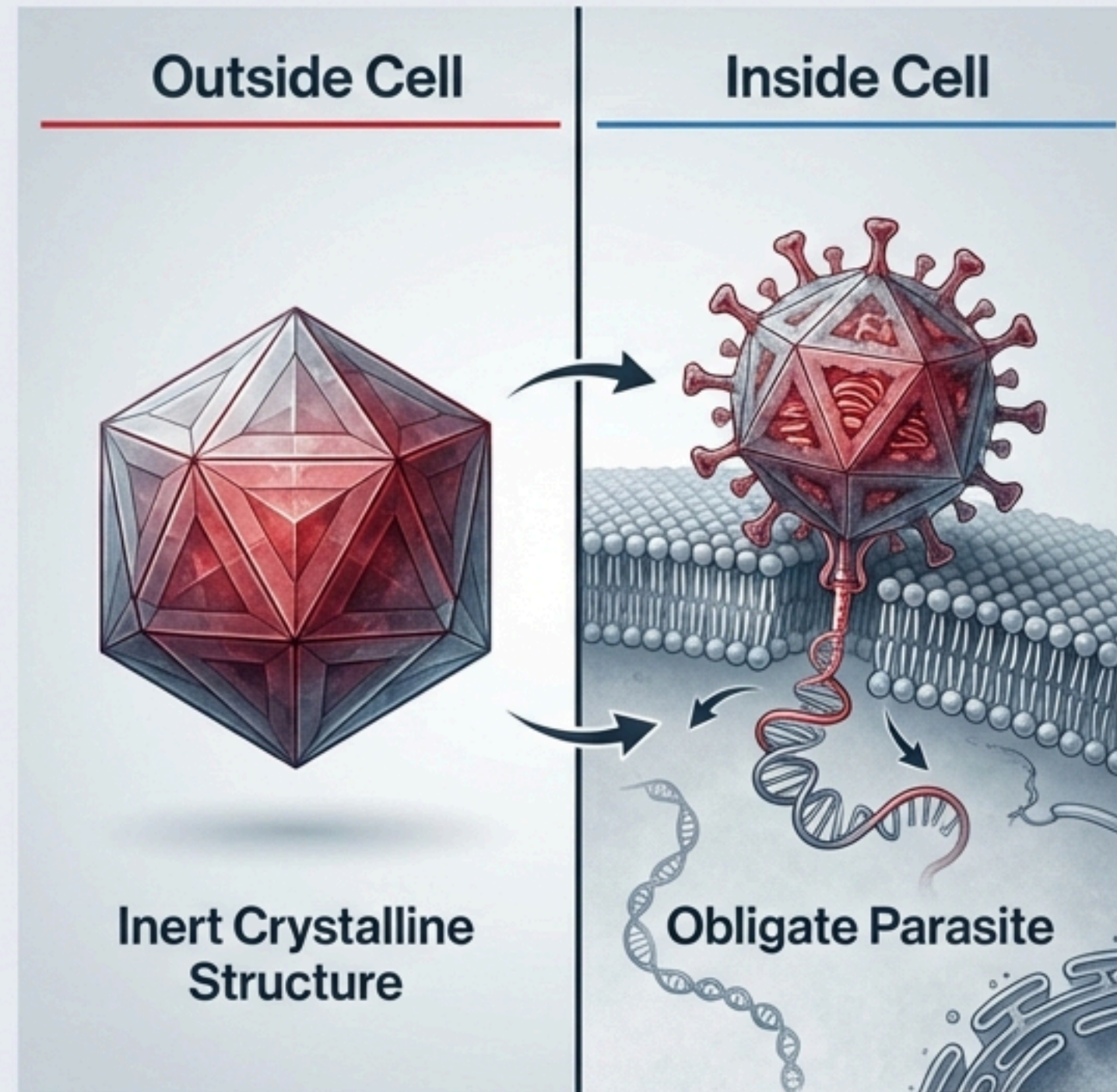
Reproduction: Sexual reproduction by copulation of male and female followed by embryological development.

The Outsiders: Viruses, Viroids, and Prions

Status: NOT included in Whittaker's Five Kingdom Classification.

Why? They are not considered truly "living" because they lack a cell structure.

Mechanism: Take over host machinery to replicate, killing the host.



“Virus means venom or poisonous fluid.”

History of Virology: A Timeline



Pasteur

Implied the term 'Virus' (Venom/Poison).



D.J. Ivanowsky (1892)

Discovered causal organism of Tobacco Mosaic Disease.

Smaller than bacteria (passed through bacteria-proof filters).



M.W. Beijerinck (1898)

Demonstrated extract of infected plants infects healthy ones.

Coined '**Contagium vivum fluidum**' (infectious living fluid).



W.M. Stanley (1935)

Showed viruses could be **crystallised**.

Crystals consist largely of proteins.

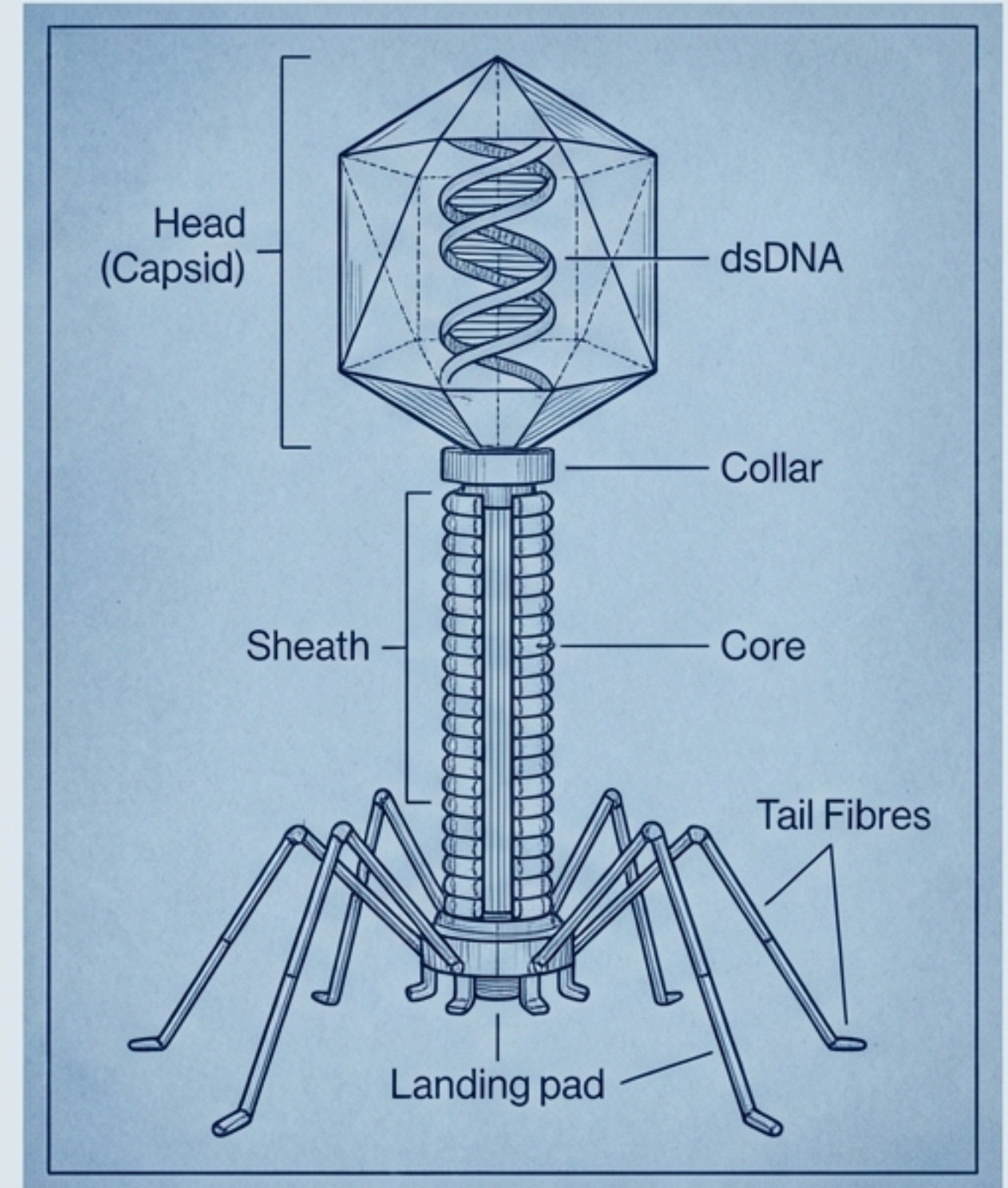
Viral Structure: The Nucleoprotein

THE GOLDEN RULE:

No virus contains both RNA and DNA.

Plant Viruses	Single-stranded RNA
Animal Viruses	Single/Double-stranded RNA OR Double-stranded DNA
Bacteriophages	Usually Double-stranded DNA

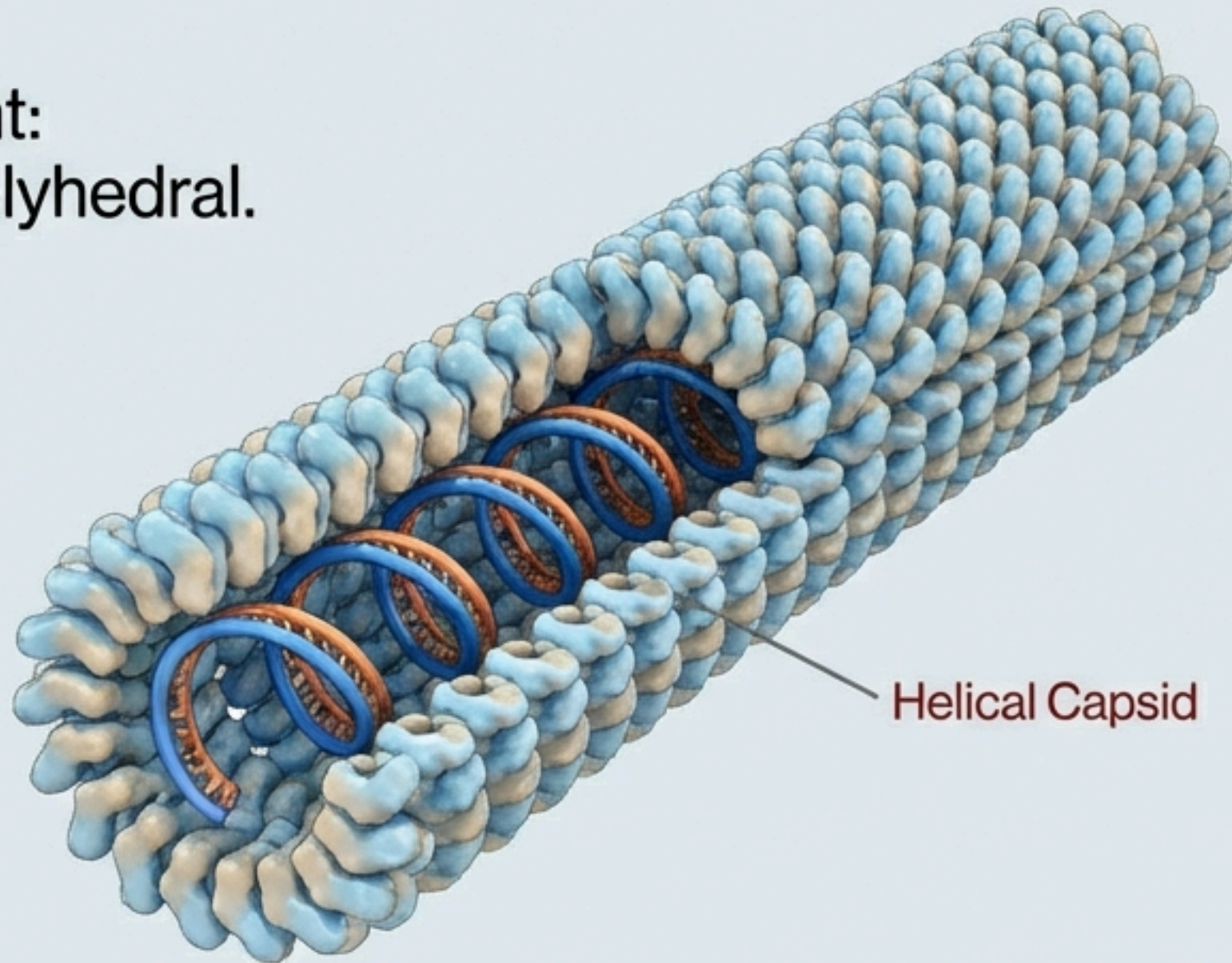
The genetic material is infectious.



Capsids & Consequences

The Protein Coat (Capsid) is made of subunits called **capsomeres**.

Arrangement:
Helical or Polyhedral.



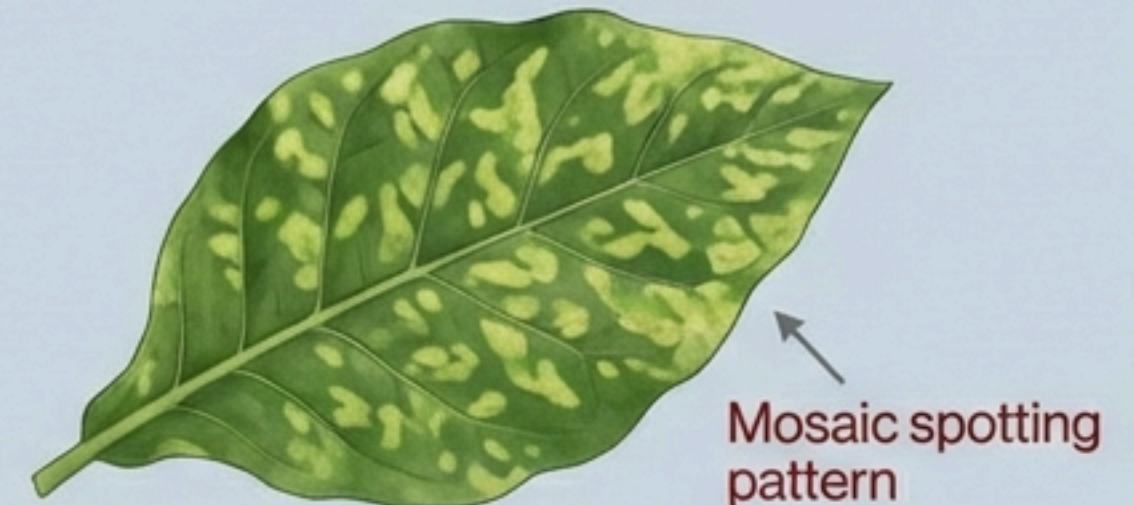
Symptoms & Diseases

Human:

Mumps, Smallpox, Herpes,
Influenza, AIDS.

Plants:

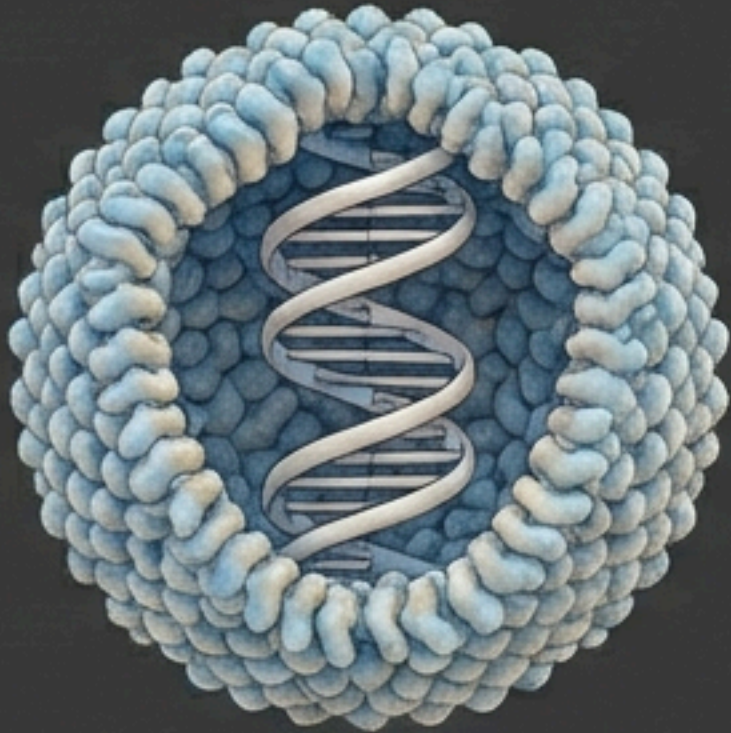
- Mosaic formation
- Leaf rolling and curling
- Yellowing and vein clearing
- Dwarfing / Stunted growth



Viroids: Smaller than the Smallest

- Discovered by **T.O. Diener** (1971).
- Disease: **Potato Spindle Tuber Disease**.

Virus: Protein + Genetic Material



Viroid: Free RNA only



Scale | _____

- Structure: **Free RNA** of **low molecular weight**.
- Lacks the protein coat found in viruses.

Prions & Lichens

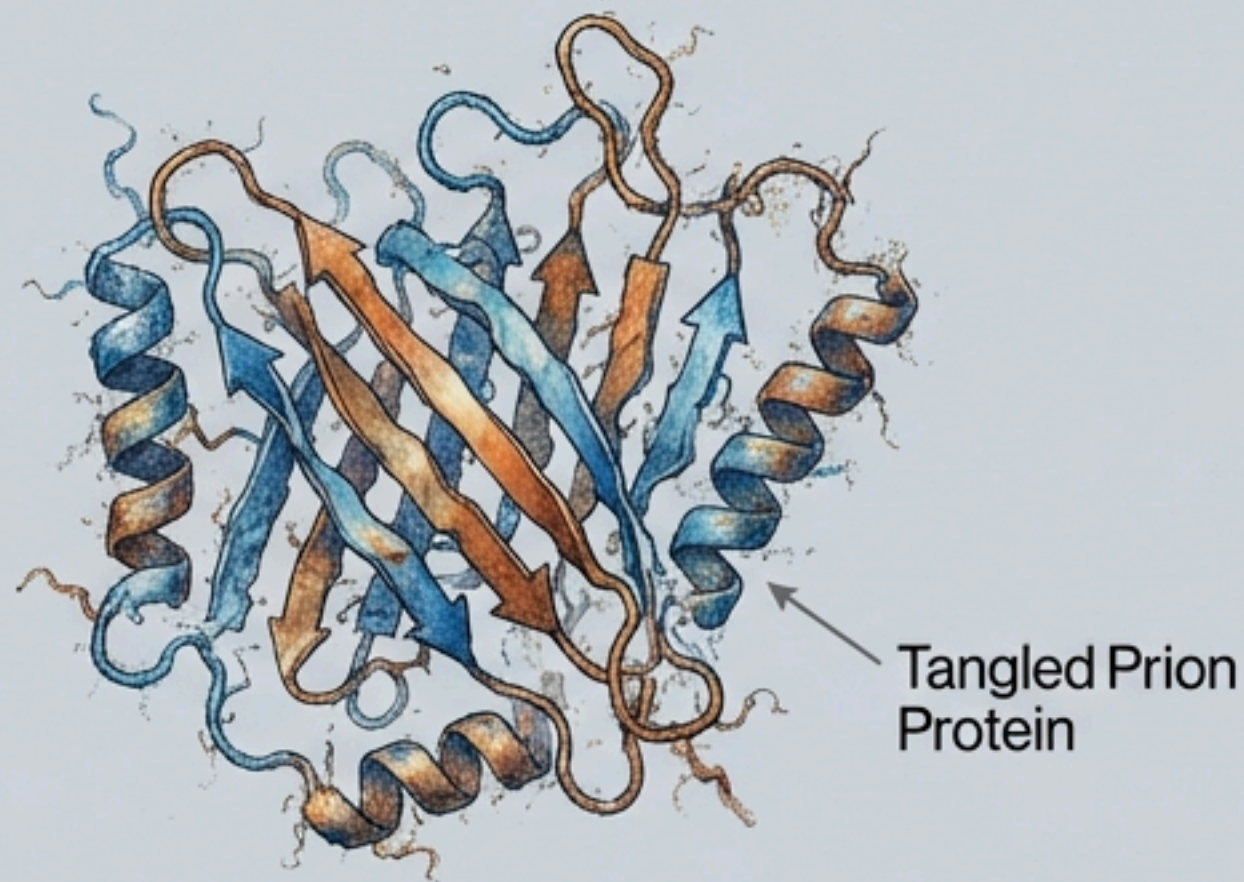
Prions (The Twisted Proteins)

Agent: Abnormally folded protein. Similar size to viruses.

Causes infectious neurological diseases.

Examples:

- **BSE** (Mad Cow Disease) in cattle.
- **Cr-Jacob Disease (CJD)** in humans.

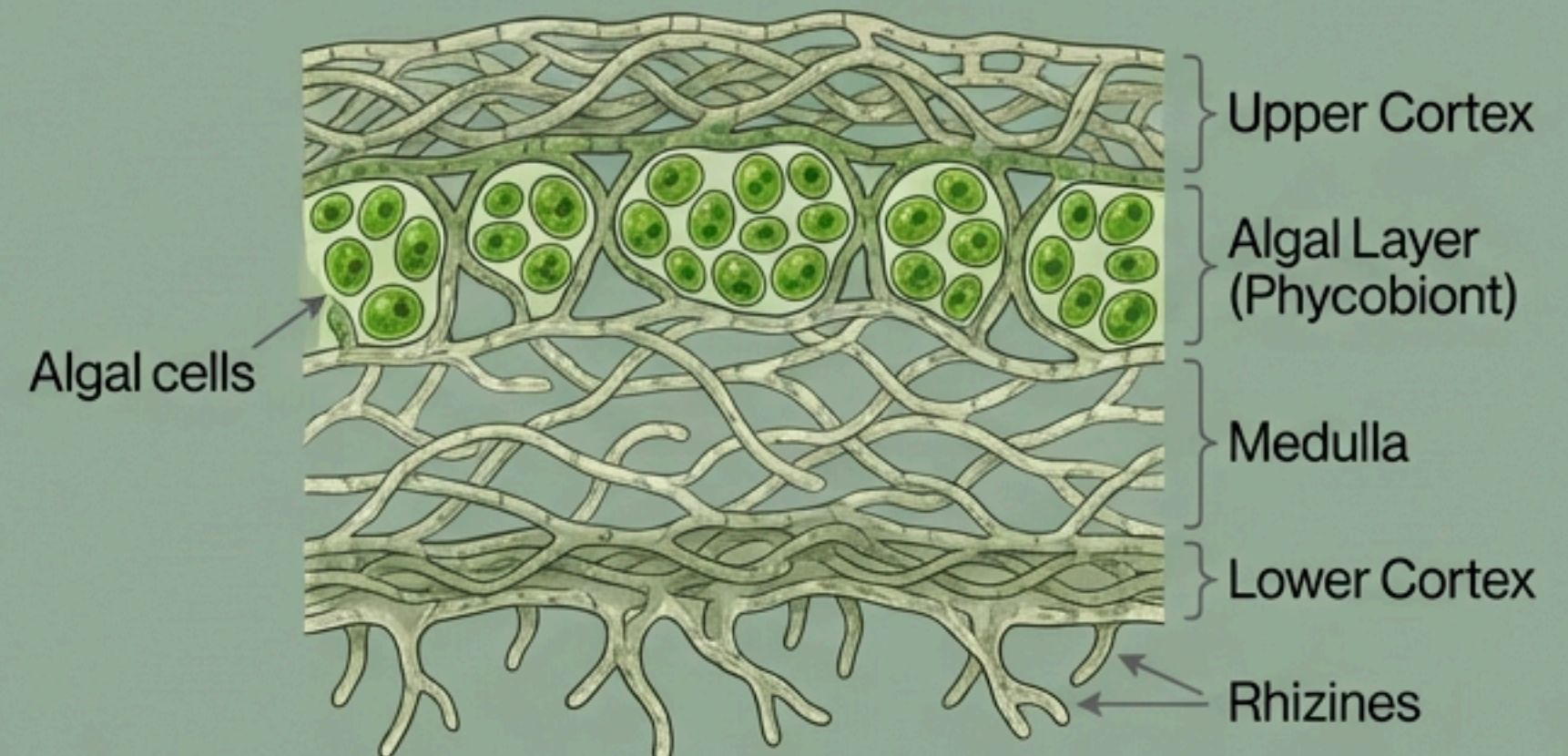


Lichens (The Symbionts)

Symbiotic association: Algae + Fungi.

- **Phycobiont** (Algae): Autotrophic, prepares food.
- **Mycobiont** (Fungi): Heterotrophic, provides shelter & nutrients.

Pollution Indicators: Do not grow in polluted areas.



Rapid Revision: Comparative Analysis

Feature	Plantae	Animalia	Virus	Viroid
Cell Type	Eukaryotic	Eukaryotic	Acellular	Acellular
Cell Wall	Cellulose	Absent	Protein Capsid	None
Genetic Material	DNA	DNA	DNA or RNA	Free RNA (Low MW)
Nutrition/ Nature	Autotrophic (mostly)	Heterotrophic (Holozoic)	Obligate Parasite	Infectious Agent

The Exam Gauntlet: Level 1

Q1. Statement Analysis: Which statement is correct regarding viral genetics?

- A) They contain both RNA and DNA.
- B) No virus contains both RNA and DNA.
- C) Plant viruses always have DNA.
- D) Bacteriophages always have RNA.

Q2. Match the Scientist:

(A) Ivanowsky (B) Beijerinck (C) Stanley (D) Diener

Matches: (i) Crystallisation (ii) **Contagium vivum fluidum** (iii) TMV discovery (iv) Viroid

Q3. Assertion (A) / Reason (R):

A: Lichens are very good pollution indicators.

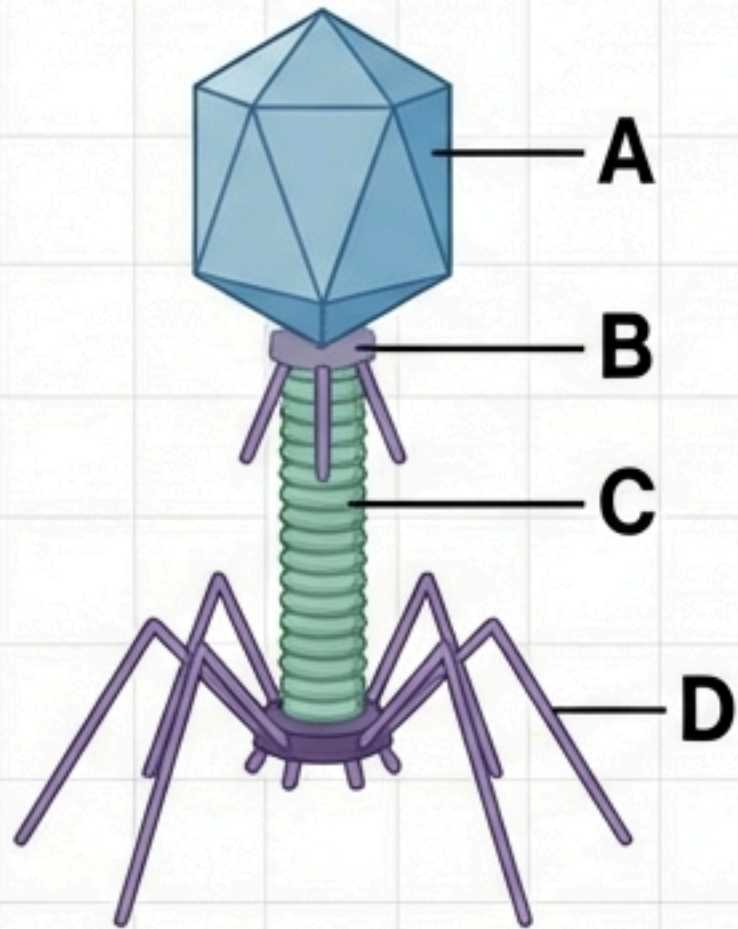
R: They do **not** grow in polluted areas.

Q4. True or False: “Viroids have RNA of high molecular weight.”

Q5. Identify: “I am a partial heterotroph that traps insects.” (Options: Cuscuta, Bladderwort, Mushroom, Algae).

The Exam Gauntlet: Level 2

Q6. Visual Labeling:



A: Head
B: Collar
C: Sheath
D: Tail Fibres

Q7. Statement Correction: The original statement is presented:
Animal viruses always contain double-stranded DNA.
(Correction: Animal viruses can contain double-stranded or single-stranded DNA or RNA.)

Q8. Concept Link: What is the human analogous variant of Bovine Spongiform Encephalopathy (Mad Cow Disease)?
Answer: Creutzfeldt-Jakob disease (CJD)

Q9. Classification: Which kingdom is characterized by food storage as glycogen or fat and a lack of cell walls?
Answer: Animalia

Q10. Analysis: "Viruses are obligate parasites that are inert outside a host cell." (True / False).
Answer: True

Answer Key & Explanations

Level 1

1. **(B)** No virus contains both RNA and DNA.
2. A-(iii), B-(ii), C-(i), D-(iv).
3. **Both A and R are true**, and R is the correct explanation.
4. **False.** Viroids have *low* molecular weight RNA.
5. **Bladderwort** (or Venus Fly Trap).

Level 2

6. A: Head, B: Collar, C: Sheath, D: Tail Fibres.
7. **Correction:** Animal viruses have single/double-stranded RNA *or* double-stranded DNA.
8. **Cr-Jacob Disease (CJD).**
9. **Kingdom Animalia.**
10. **True.**

Master the NCERT lines to master the Exam.